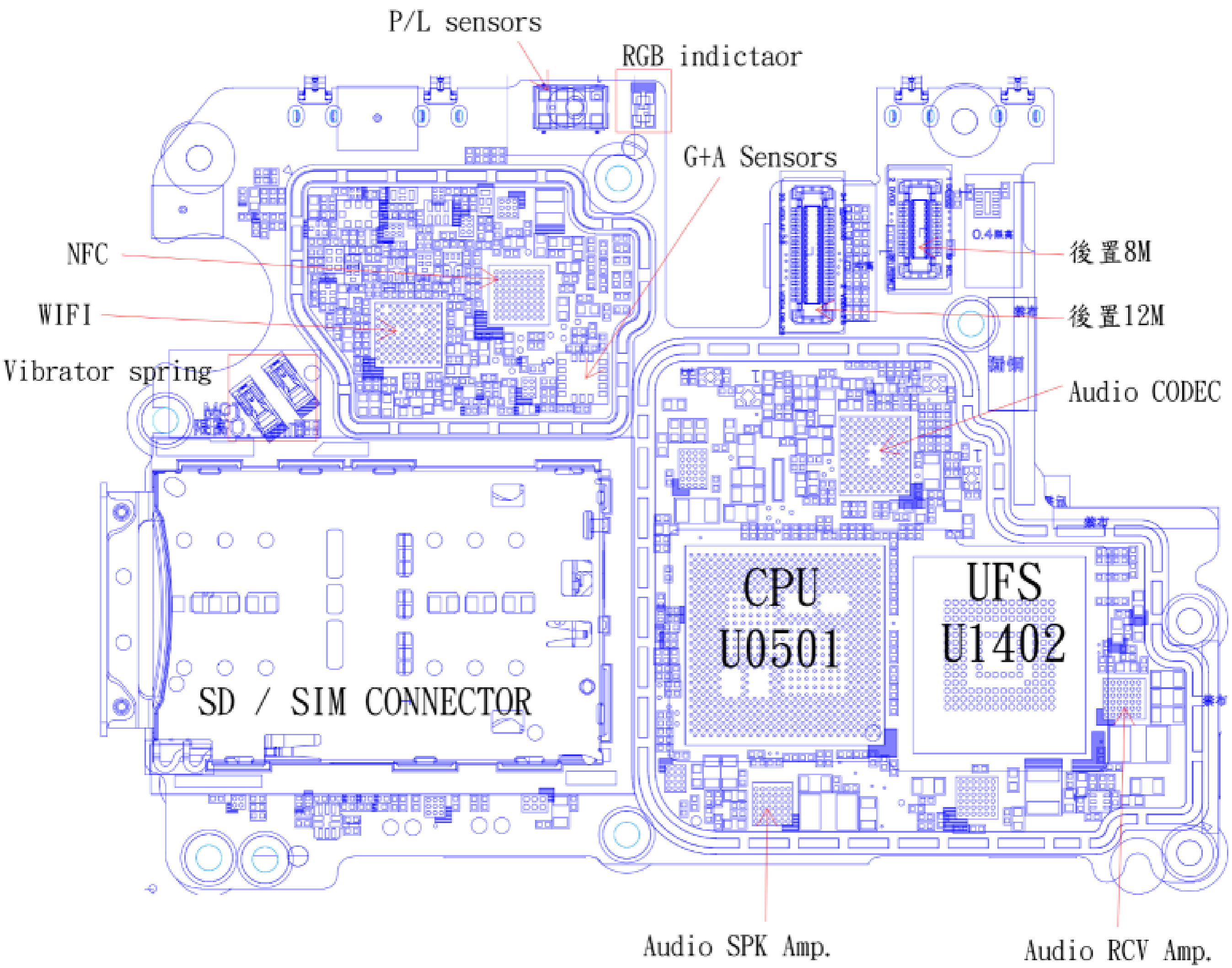
Draco ZS620KL Troubleshooting Guide

2018.04.30 H Huang

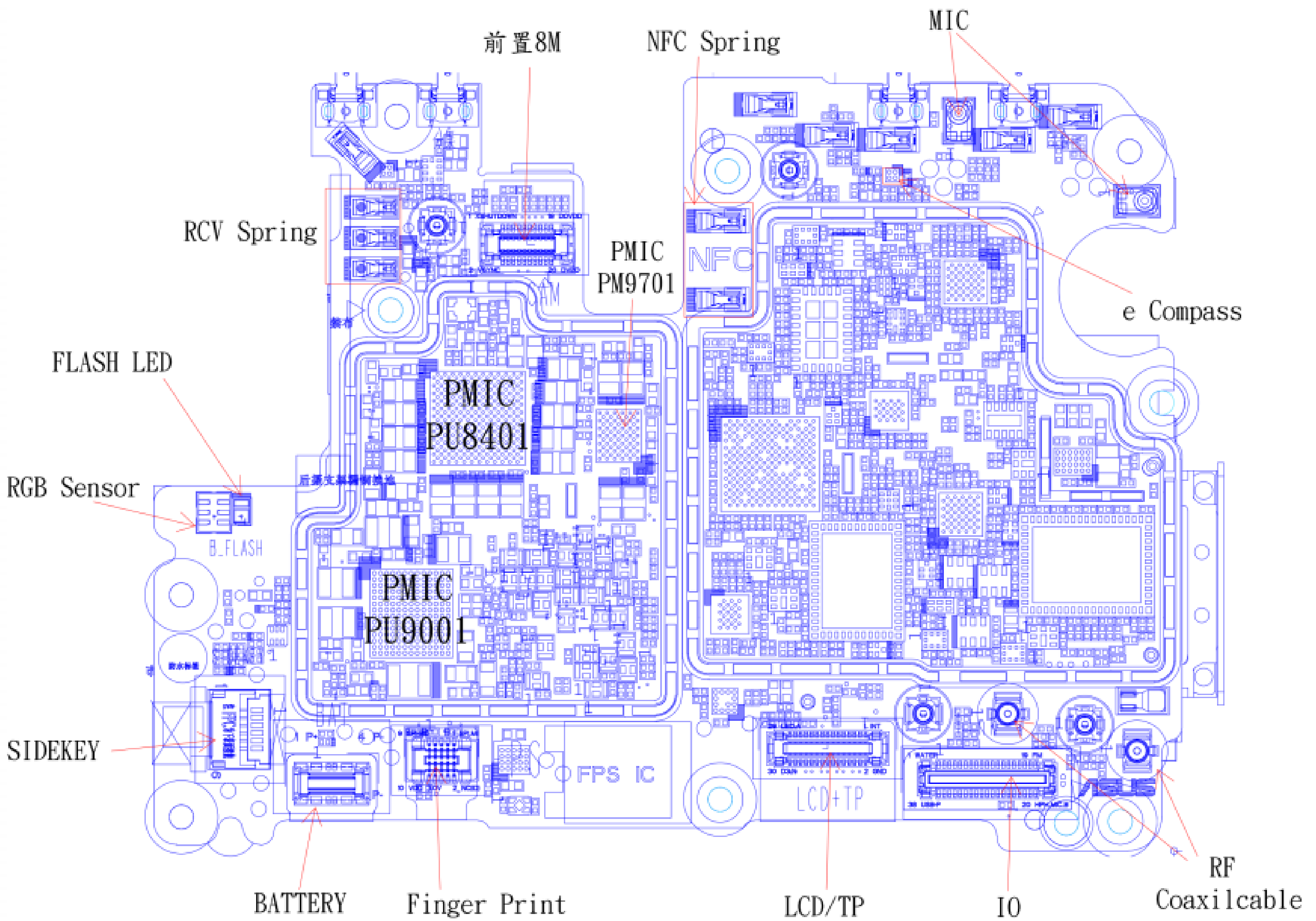
Outline

- Component and PCB Board Introduction
- Power On Sequence
- Troubleshooting Guide

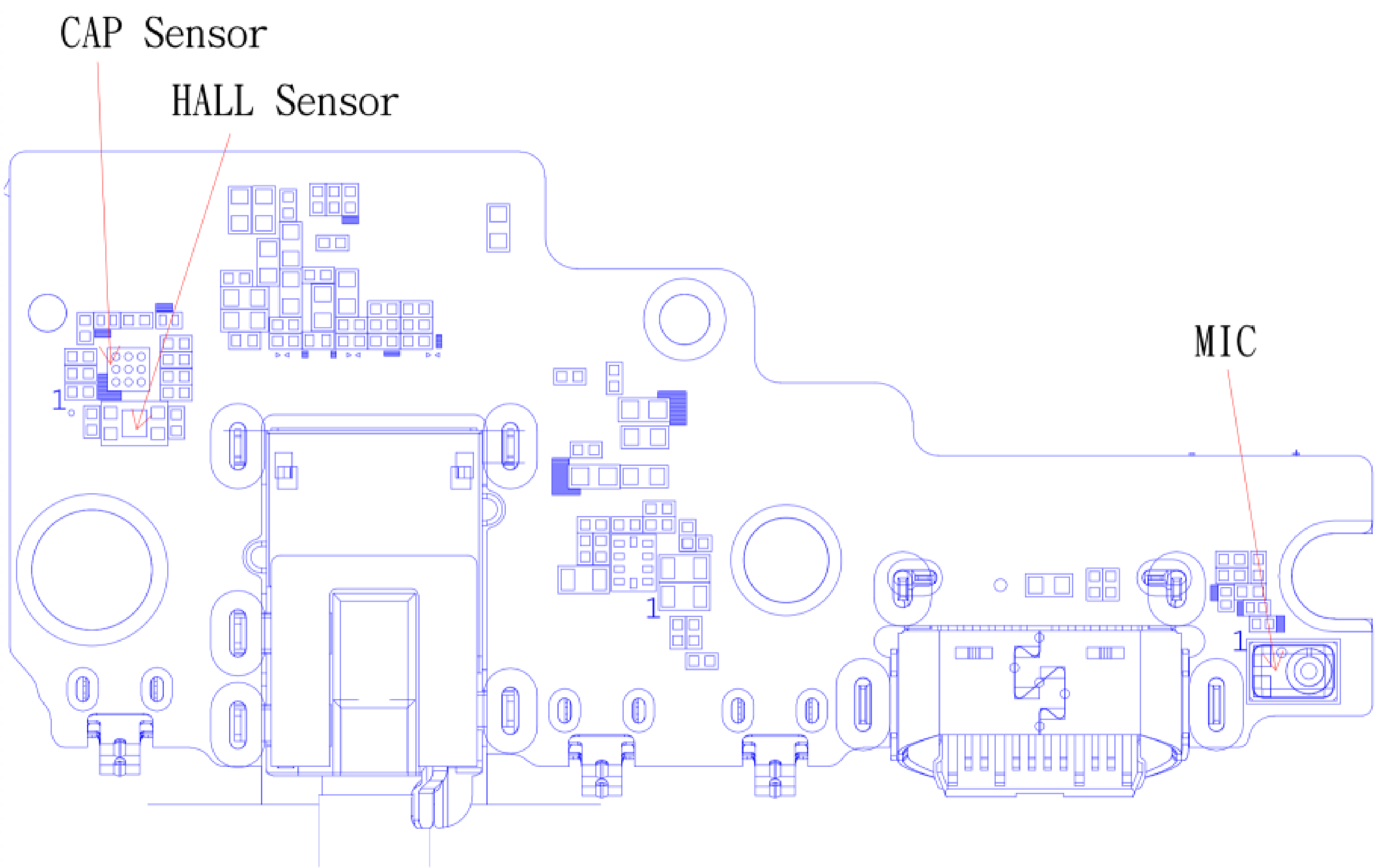
MB TOP


P/L sensors
RGB indictaor
G+A Sensors
NFC
WIFI
Vibrator spring
後置8M
後置12M
Audio CODEC
CPU
U0501
UFS
U1402
SD / SIM CONNECTOR
Audio SPK Amp.
Audio RCV Amp.

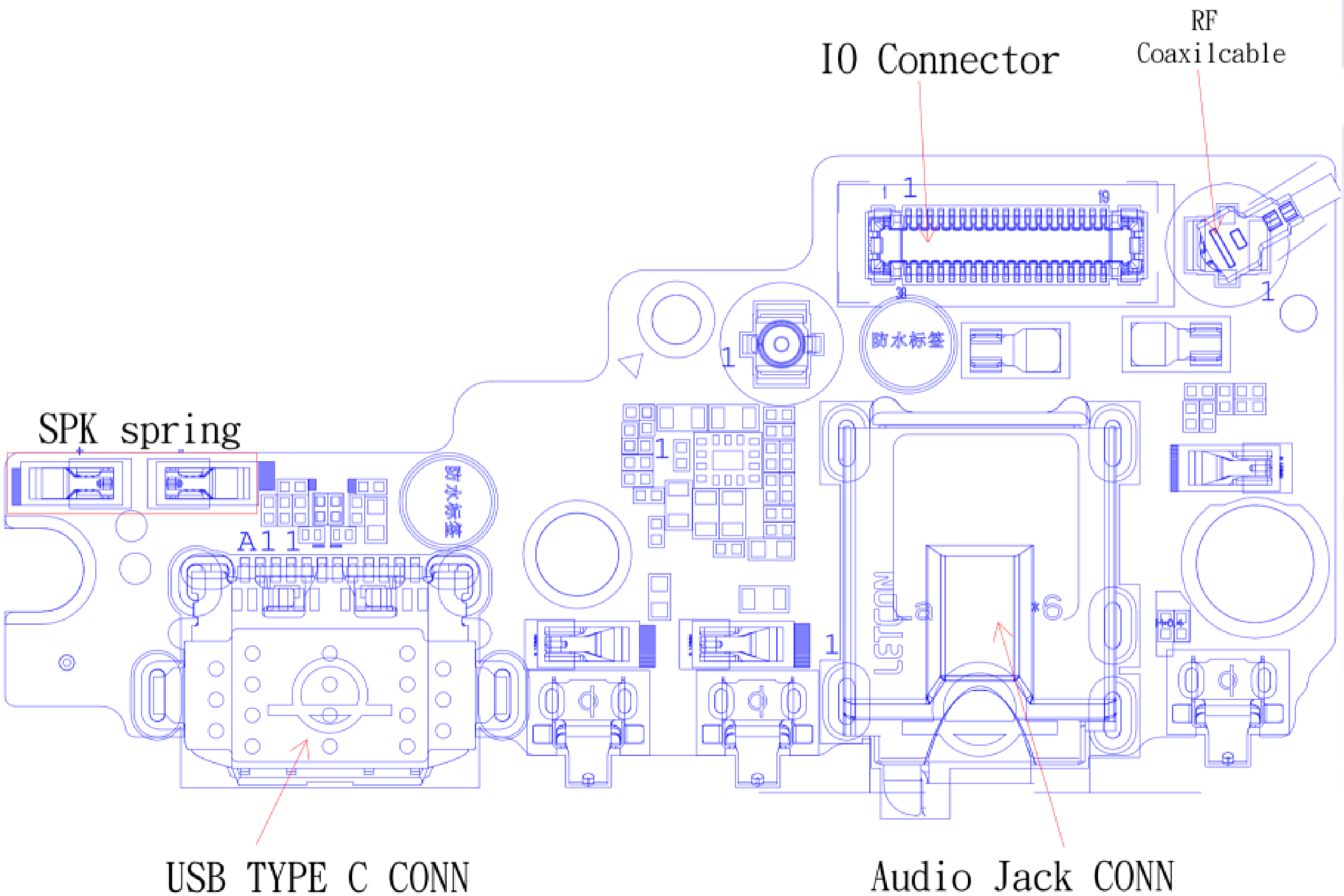
MB Bottom


前置8M
NFC Spring
MIC
RCV Spring
PMIC
PM9701
e Compass
FLASH LED
PMIC
PU8401
RGB Sensor
B_FLASH
PMIC
PU9001
SIDEKEY
FPS IC
LCD+TP
BATTERY
Finger Print
LCD/TP
IO
RF
Coaxilcable

KB TOP



KB Bottom


IO Connector
RF
Coaxilcable
SPK spring
防水标签
USB TYPE C CONN
Audio Jack CONN

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POWER ON SEQUENCE (1)

Power Sequence	Location	Boot Votlage	Functional description
1. PON Trigger	T8716.1	1.8	Power Button Trigger = 0V
2. REF BYP	C8703.1	1.25	PM 845 internal reference
3. PON OUT	T8703.1	1.2	PMI/ PM8005 Power on trigger
4. VREG S5	PC8511.1	1.904	For MV subregulation
5. VREG XO	C8802.1	1.8	XTAL used
6. VREG BOB	PC9104.1	3.4~3.7	PMI8998 BOB
7. VREG S7	PC8514.1	1.04	For LV subregulation
8. VREG S6	CE1205.1	0.928	SoC MX
9. VREG L4	C1217.1	0.928	LPI MX
10. VREG S1	CE1102.1	0.928	EBI PHY
11. VREG S9	CE1204.1	0.872	SoC CX
12. VREG L27	PC8615.1	0.872	LPI CX
13. VREG S3	CE1103.1	1.352	For LV subregulation
14. VREG L11	PC8610.1	1	RF WTR 1.0 V digital
15. VREG S4	T2101.1	1.8	System I/O and digital
16. VREF MSM	C1215.1	1.2	SoC SD/UIM reference
17. VREG L7	R4202.1	1.8	RF WCN XO
18. VREG L5	PC8608.1	0.8	RF WCSS MX/ CX
19. VREG L17	PC8613.1	1.304	RF WCSS ADC, WCN
20. VREG L25	PC8606.1	3.104	RF WCN 3.3 V channel 0

POWER ON SEQUENCE (2)

21. SLEEP_CLK	T0502.1	1.8	32 kHz sleep clock
22. RF_CLK2	C6116.2	0.8	38.4 MHz RF XO clock output
24. VREG_L26	PC8614.1	1.2	PHY (DSI, CSI)
25. VREG_L1	C1127.1	0.88	PHY UFS, USB)
26. VREG_L12	PC8612.1	1.8	PLL, BBCLK, PHY (USB, UFS)
27. LN_BB_CLK1	R8806.2	1.8	19.2 MHz RF XO clock output
31. VREG_S2C (PM8005)	CE1016.3	0.6	MODEM
28. VREG_S2	CE1013.1	1.128	LPDDR4_VDD2
32. VREG_L24	C1120.1	3.088	USB
29. GPIO_26	T8711.1		Single line bus handshake for PM8005
30. VREG_S3C (PM8005)	C1402.1	4	
33. VREG_L20	CE1104.3	2.96	UFS
	C1209.1	0.752	LPDDR4X VDDQ
34. VREG_L2	1	1.2	UFS
35. VREG_L13	PC8617.1	2.96	WTR 1.0 V analog
36. VREG_L21	PC8604.1	2.96	SD/MMC card
37. VREG_S13	CE1005.3	0.872	Kryo Silver
38. PON_RESET_N	T8702.1	1.8	Power-on reset output signal
38. PS_HOLD	T8704.1	1.8	Power-supply hold signal to PMIC

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Power on failure

- **Check LCM module and battery contact well**

- **Check before power on**

1. Check VBAT (T9602.1) >3.4V , VPH_PWR (PC8408.1) > 3.4V, if 0V check maybe somewhere short to GND
2. Check PHONE_ON_N voltage, T2103.1 = 1.8V, if not re-solder or change PU8401
3. Check FAULT_N , T8706.1 voltage is 1.2V or not, if not re-solder or change PU8401/ PU9001/ PU9701

- **Check within pressing power key**

1. Check PHONE_ON_N_CON (T8716.1) be low, when push power button, if not check CON2101, R2103

- **Check after pressing power key**

1. Check MSM_PS_HOLD (T0501.1) is Keep high =1.8V
2. If can recognize USB port but can not download, check U1401/ U1402
3. If can not recognize USB port
   1. Change KB brd. or IO FPC assemble
   2. Check the USB relativity component (USB SW PU8301, U0501)
4. If MSM_PS_HOLD (T0501.1) not high
   3. check REF_BYP_PM845 (C8703.1) voltage ~ =1.22V, if not then check PU8401
   4. check PM8005_REF_BYP (C9802.1) voltage ~ =1.22V, if not then check PU9701
   5. confirm all power on sequence relate power rail is normal
5. Check XTAL (X8801) is work

Display failure

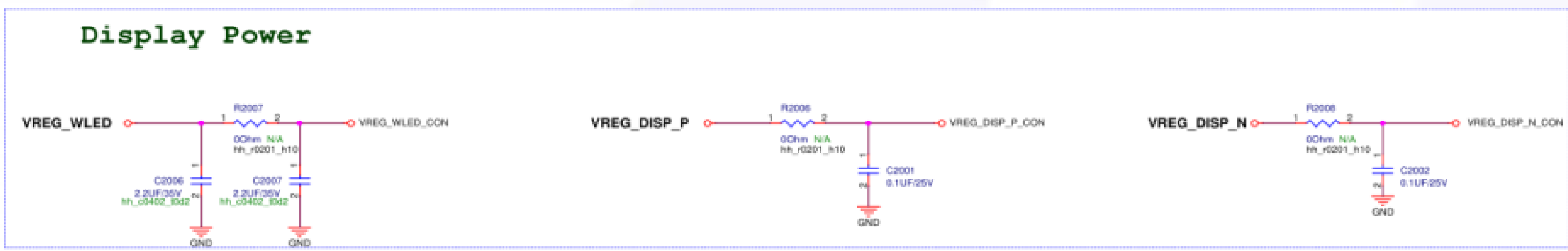
Display blackout after a flicker, or just no display while ADB device can be recognized after booting up.

1. Check CON2001 for any short circuits, shift, or solder skips.
2. Swap the testing panel and also check the connector on the panel FPC.
3. Check VPH_PWR > 3.4V
4. CheckVREG_WLED_CON > 20V

   if not, checkVREG_DISP_PD9200, PL9200, R2007
5. VREG_DISP_P_CON/ VREG_DISP_N_CON =+/-5.5 V

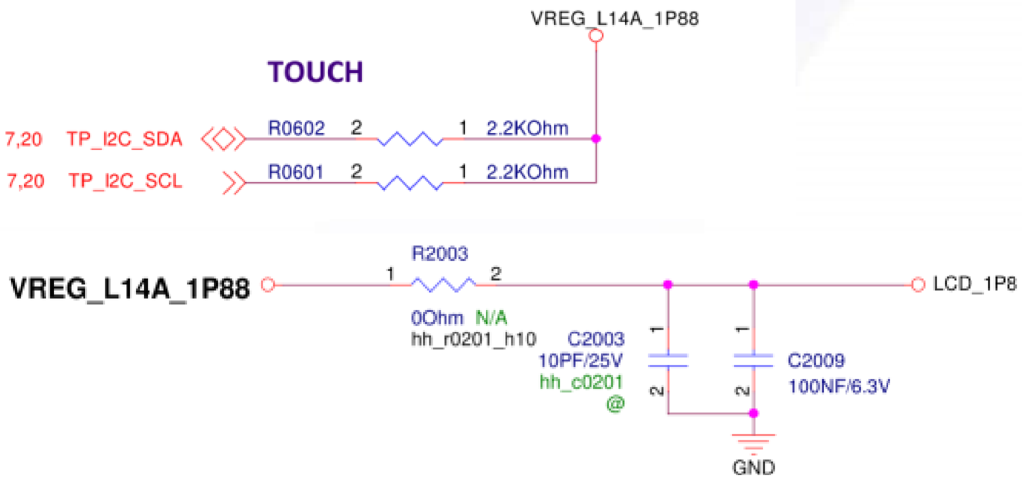
   if not, check PL9202/ PL9203 / R2006/ R2008 for cracks, short circuits, wrong polarity.
6. Check LCD_1P8 = 1.8 V

   if not, check R2003
7. Check WLED sink side, L2001/ L2002
8. Check LCD_RESET_N (D2001.1) = 1.8 V



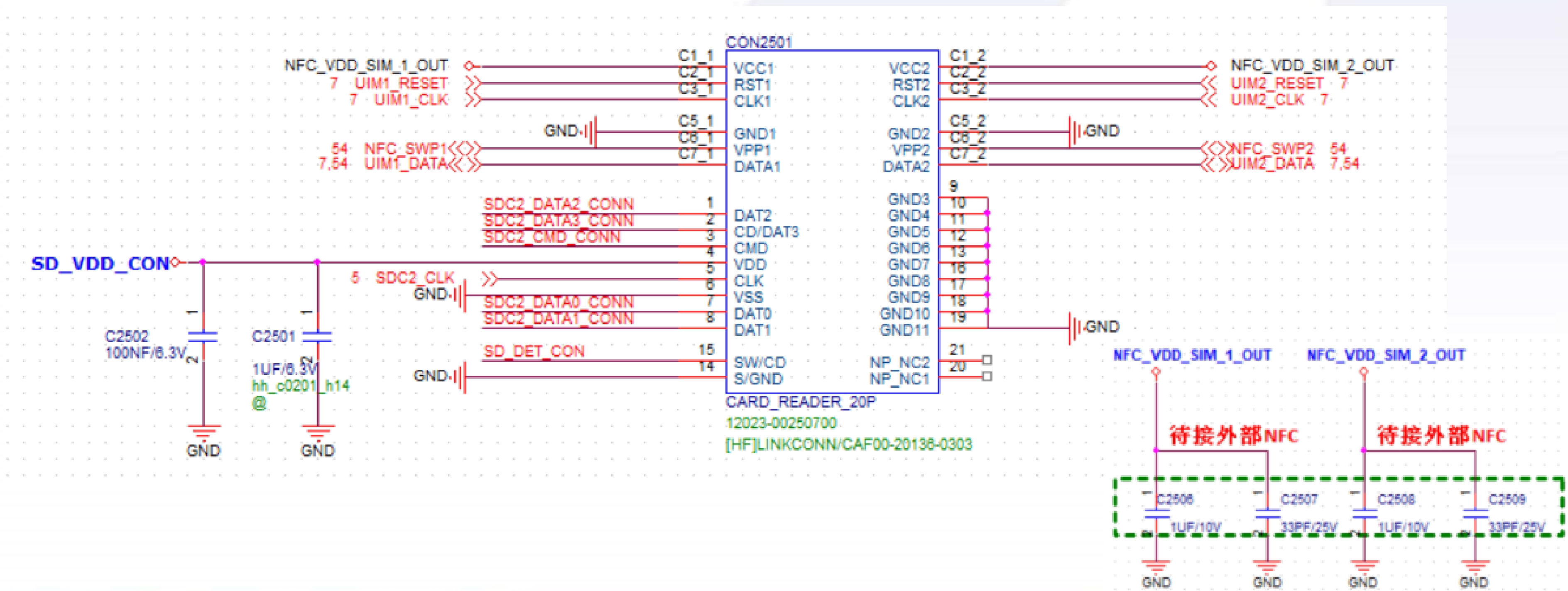
Touch failure

1. Check CON2001 for any short circuits, shift, or solder skips.
2. Swap the testing panel and also check the connector on the panel FPC.
3. Check LCD_1P8 = 1.8 V
4. Check R2003 for cracks, short circuits, wrong polarity.
5. Check below signal:
   - ◎ TS_RESET_N (T2014.1) = 1.8 V
   - ◎ TP_I2C_SDA/ TP_I2C_SCL has pull high , R0601.2/ R0602.2 = 1.8V
   - ◎ TS_INT_N has any high-low transition when touch screen be touched



SIM test failure

- Re-install SIM card or replace before testing it again;
- Visual check SIM tray for damage, soldering for skew or overweight;
- Measure the SIM power supply voltage NFC_VDD_SIM_1_OUT、NFC_VDD_SIM_2_OUT
- If no power input re-solder or replace U5401 NFC
- Re-solder or replace U0501 CPU



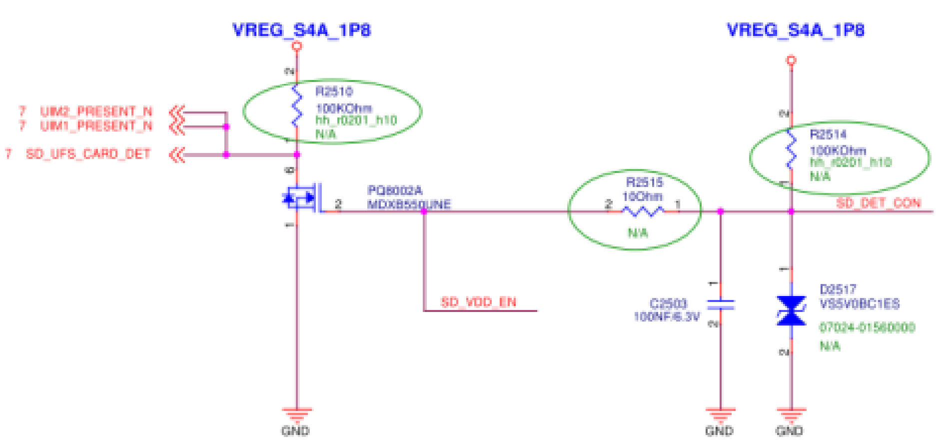
SD CARD test failure

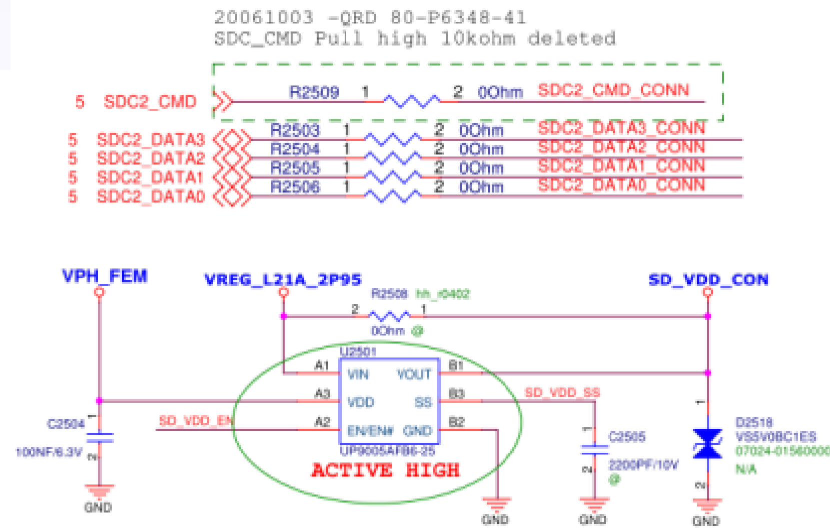
1. Check SD_UFS_CARD_DET status should be low when card insert (R2510.2 =0V)
2. Check SD_VDD_CON = 3V (R2508.1 =3V)

   if not , check SD_VDD_EN it should be high (R2515.2 =1.8V)

   if Item 1 or 2 abnormal, check R2510, R2514, R2515, U2501 for cracks short circuits, wrong polarity.
3. If SD_VDD_EN = 1.8V, but SD_VDD_CON not 3V, then change U2501
4. If has 3Voutput but can not recognize SD CARD， please check

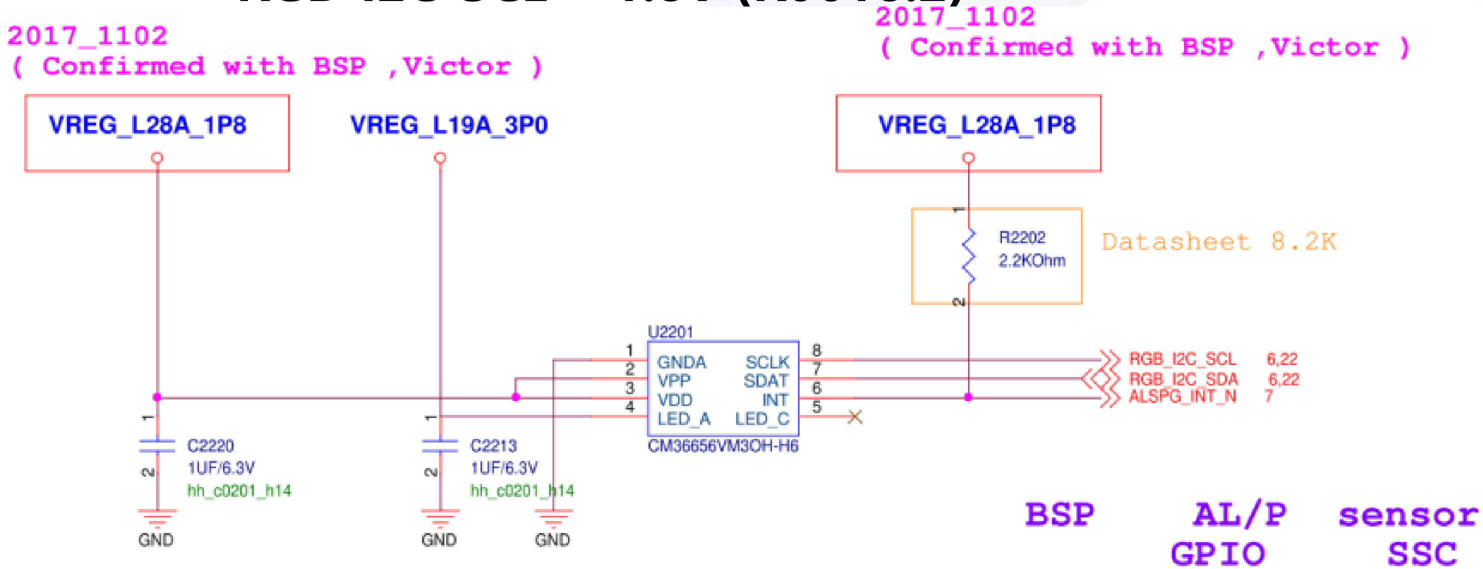
   DATA0~3/CMD/CLK (R0507/ R2503 ~ R2506/ R2509)
5. Check SD stocket for damage, soldering for skew or over height





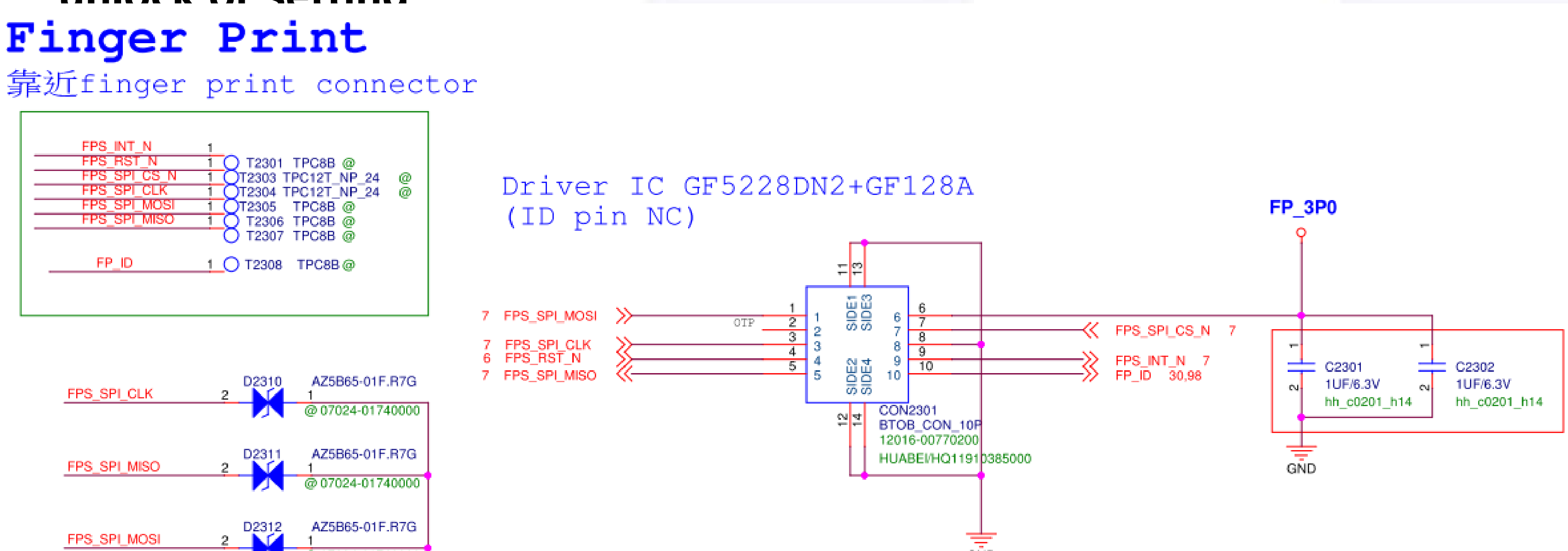
Proximity/Light Sensor failure

1. Check P/L sensor do not has anything defect blocking on glass open area
2. Check below two power source
   - ◎ VREG_L28A_1P8 = 1.8V (C2220.1)
   - ◎ VREG_L19A_3P0 = 3V (C2213.1)
3. Check I2C has pull high 1.8V
   - ◎ RGB_I2C_SDA =1.8V (R0619.2)
   - ◎ RGB_I2C_SCL = 1.8V (R0618.2)



Finger Print Sensor failure

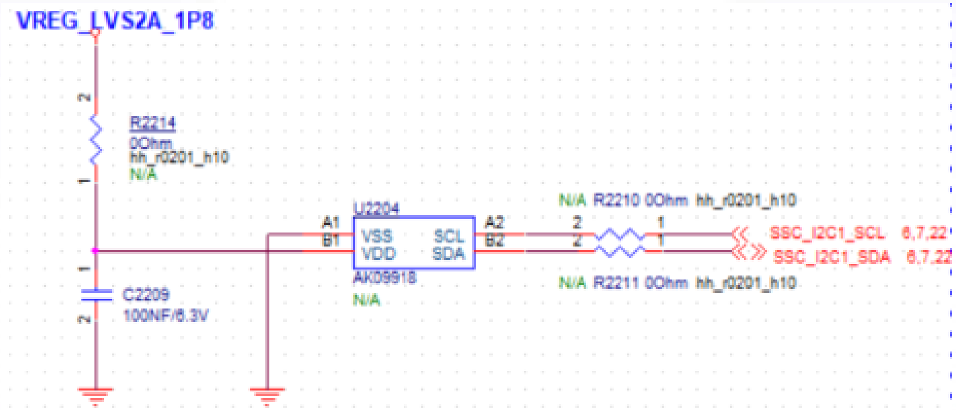
1. **Check FPC connect well, check CON2301** has any short circuits, shift, or solder skips.
2. **Swap Finger Print Sensor Module，check Module status**
3. **Check FP_3P0 =3V, if no voltage output change LDO (PU9501)**
4. **Check below singal**
   - ◎ **FPS_SPI_CS_N = 1.8V (T2303.1)**
   - ◎ **FPS_RST_N = 1.8V (T2301.1)**
5. **Check FPS_INT_N has high-low transition (T2301.1) when FP unlock or setting**



E-compass failure

U2204 E-compass can't initial, or I2C bus error. Please check:

1. Check the appearance of U2204, C2209 for cracks, short circuits, wrong polarity.
2. If no defect found, probe U2204 4 pins and compare the impedance with normal PCB. [check if any short circuits between each pin]
3. Check **below two power source**
   - ◎ **VREG_LVS2A_1P8 = 1.8V (R2214.1)**
   - ◎ **VREG_L19A_3P0 = 3V (C2213.1)**
4. Check **below two power source**
   - ◎ **SSC_I2C1_SCL =1.8V (R2210)**
   - ◎ **RSSC_I2C1_SDA = 1.8V (R2211)**
5. Swap U2204 to check the IC.



MIC failure

There are three MICs on the device, including on the SUB board U0601 (Main Mic), and also main board AU3601 (2nd Mic)、AU3603 (3rd Mic).

1. Check the appearance of all MIC for wrong polarity, shift, and solder overflow. Besides, also check surrounding components.

2. If all MIC appearance are normal, check the MIC through the hole using a microscope to check any foreign matters or cracks on the film

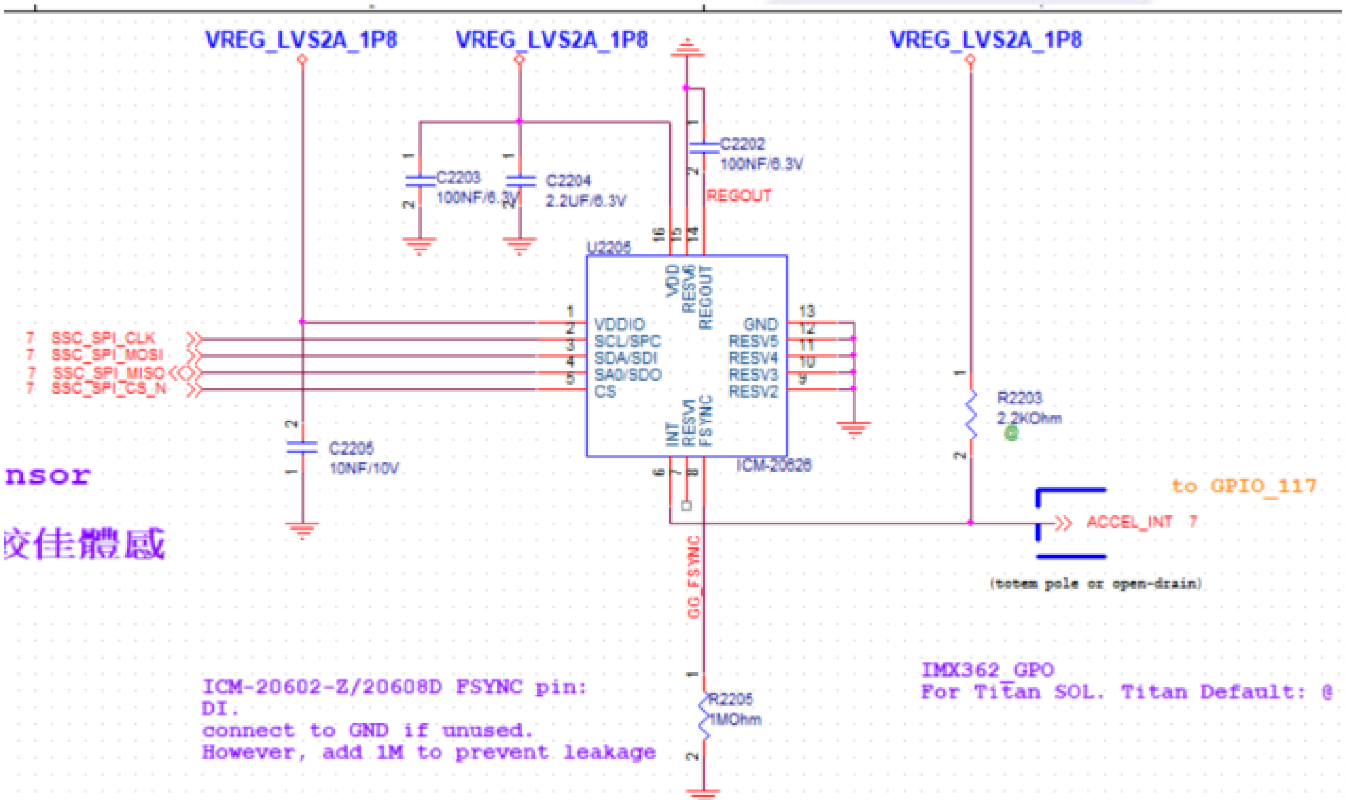
3. Add-solder or re-solder PU3501 codec, or replace it before testing again

Motor failure

- Measure the voltage on the PAD2601 and PAD2602 compare against normal MB
- Make sure vibrator contact well with sprint
- replace motor before testing it again
- Re-solder power management chip PU9001 or replace it

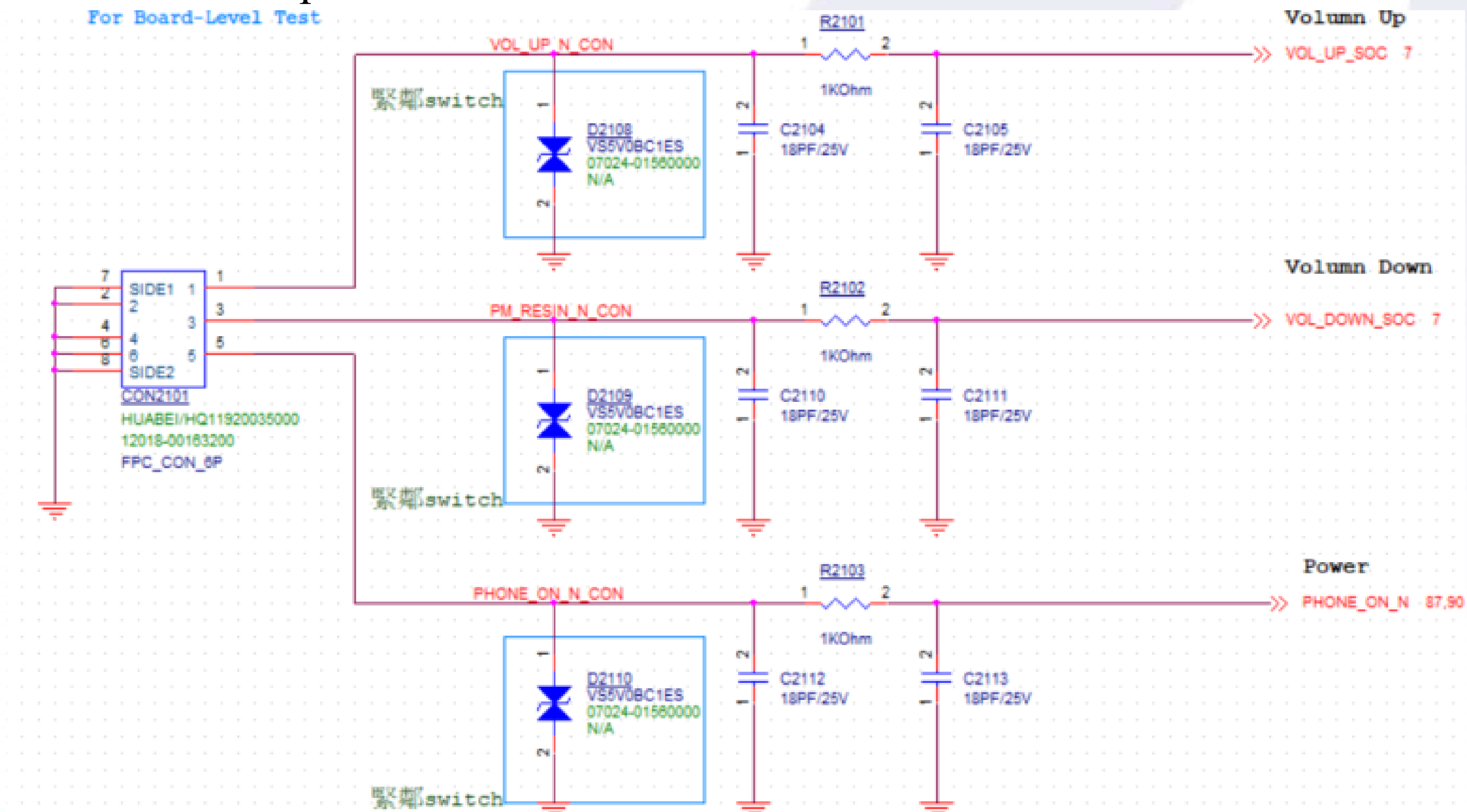
G-sensor failure

1. Measure whether the VREG_LVS2A_1P8 voltage is 1.8V
2. Re-solder or replace gravity sensor U2205
3. Check gravity sensor module circuit components for soldering errors or damage;
4. Re-soldering or replace CPU



Key failure

1. Re-install connector or replace side key before testing again;
2. Check whether POWER KEY/KYPDPWR_N voltage signal is normal (1.8V)
3. Check components on key circuit for damage or soldering errors;
4. Re-solder or replace the CPU or PMIC



Charging failure, USB identification failure

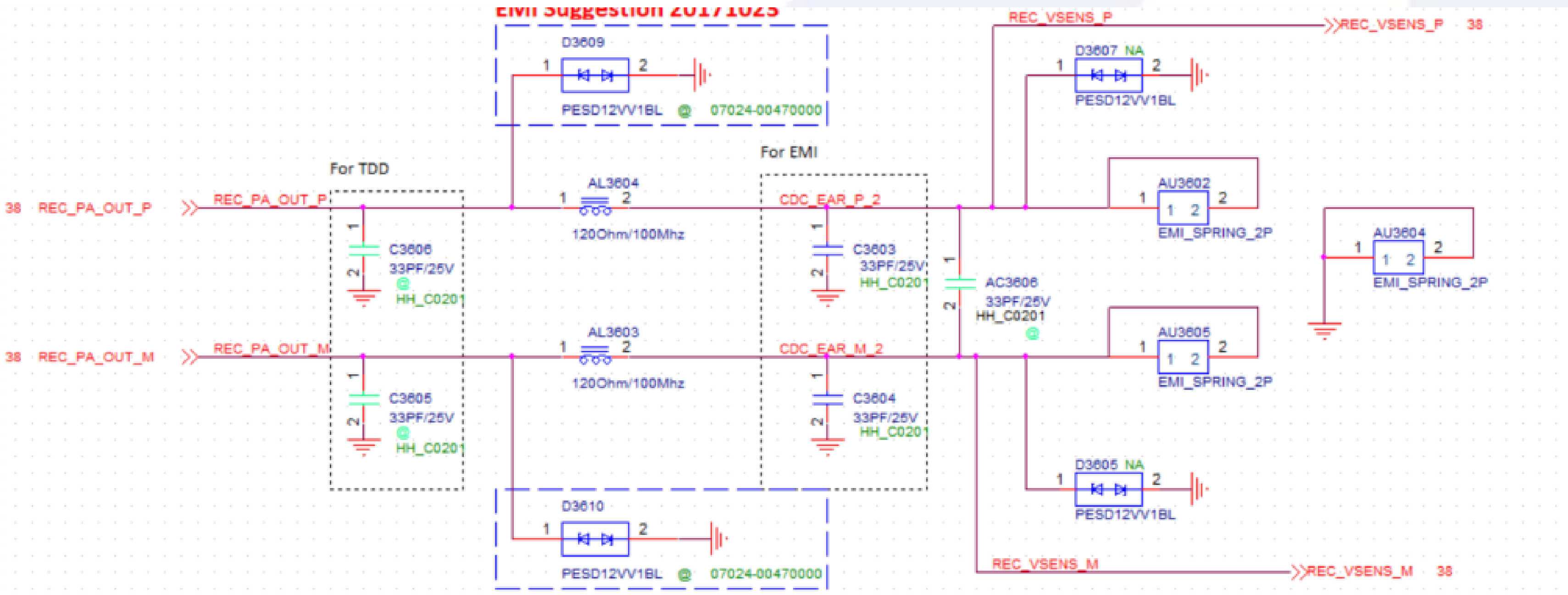
1. Re-connect KB board or replace KB board before testing it
2. Check MB's KB board connector for dirt or damage
3. Replace battery before testing again
4. Check battery connector CON9601 for dirt, bubble soldering, damage
5. Check charging management chip PU9001 for soldering error
6. Re-solder PU9001 power management chip and check for soldering errors
7. Re-solder or replace the CPU

Flashlight failure

1. Add-solder or replace flashlight LED2403
2. Re-solder or replace PMIC PU9001

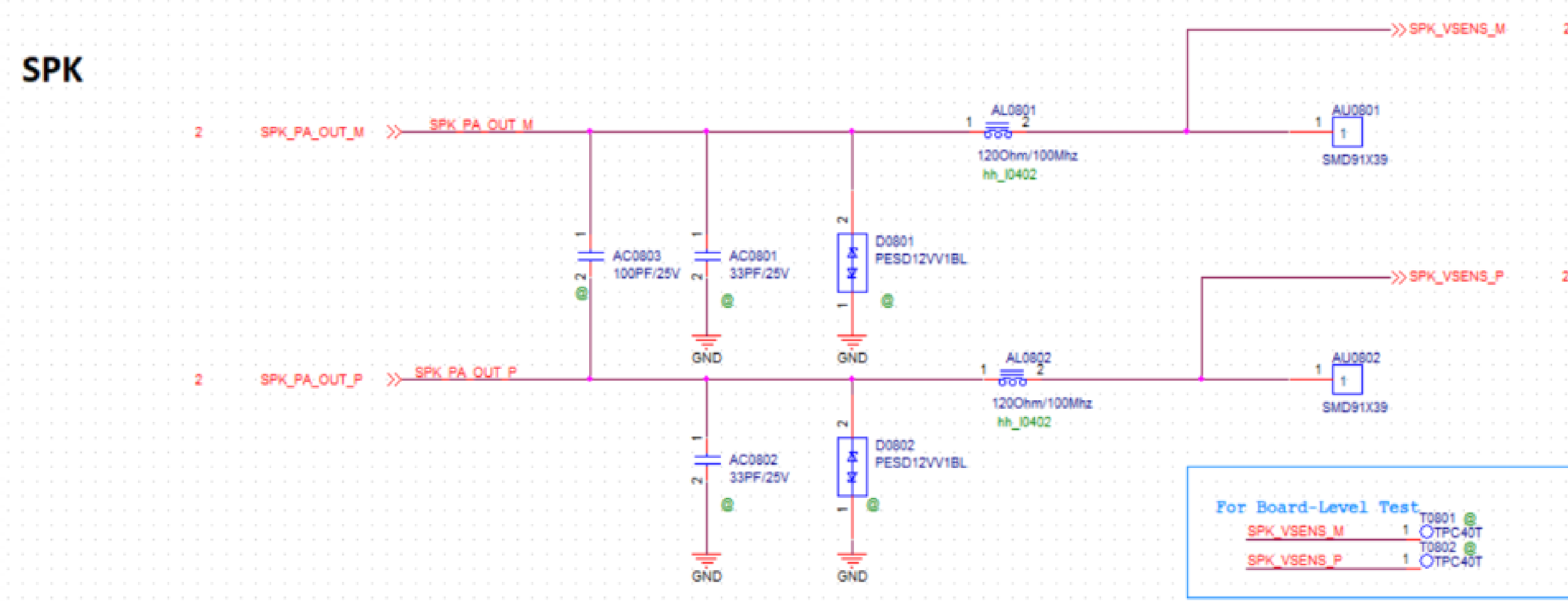
Earpiece silent

1. Re-install or replace earpiece before testing it again
2. Visually check headset circuit components for soldering offset or body damage
3. Add-solder / re-solder the audio smart amplifier AU3803 before testing it again.
4. Add-solder or replace Codec PU3501



Speaker silent

- The speaker is located on KB board. Replace KB board to test and determine whether KB board or the MB is defective and then repair it
- Visual check relevant components for damage or soldering error
- Check connector between MB and KB board for skew and continuous tine;
- Add-solder or re-solder AU3804 Audio PA or replace it;
- Add-solder or replace Codec PU3501



Camera failure

1. **Check FPC contact well , connector** has any short circuits, shift, or solder skips.
2. **Change Camera Module**
3. **Check below table voltage is normal**
   - **a. check every voltage on resistor is normal, if not check b, c**
   - **b. check LDO enable pin is high**
   - **c. check LDO SMT or change LDO**

	Net	Voltage	LDO enable	LDO	Resistor	
Rear 12M	MCAM0_DOVDD_T_1P8	1.8V	Internal	Internal	R1704	Sensors
	MCAM0_AVDD_T_2P8	2.8V	X (沒留測點)	PU9504	R1702	
	MCAM0_DVDD_T_1P05	1.05V	PT9501.1	PU9510	R1707	
	MVCM_2P8	2.8V	X	PU9503	R1712	OIS
	OIS_AVDD	2.8V	Internal	Internal	R1713	
Rear 8M	MCAM1_DOVDD_W_1P8	1.8V	X	Internal	R1709	Sensors
	MCAM1_AVDD_W_2P8	2.8V	X	PU9504	R1703	
	MCAM1_DVDD_W_1P2	1.2V	PT9502.1	PU9513	R1706	
Front 8M	CAM_DOVDD_F_1P8	1.8V	Internal	Internal	R1801	Sensors
	CAM_AVDD_F_2P8	2.8V	X	PU9504	R1803	
	CAM_DVDD_F_1P2	1.2V	PT9502.1	PU9513	R1802	

Thanks for your attention